GOOD TRUNCATIONS FOR COMPUTING INTEGRAL BASES

Abstract. We provide good truncation bound that speed up the computation of integral basis.

1. Introduction

Integral basis are very useful in real life.

Given a polynomial $f \in K[X, Y]$ monic in Y, we wish to compute the local contribution to the integral basis at the origin or the local integral basis at the origin.

2. The determinacy

We begin with some standard defitinions (see [2] for more details).

Definition 2.1. *We say that two power series $f, g \in K[[x, y]]$ are* right equivalent *if and only if there is an automorphism $\varphi \in \mathrm{Aut}(K[[x, y]])$ such that $f = \varphi(g)$, and we denote this by $f \sim_r g$.*

We call $f, g \in K[[x, y]]$ contact equivalent *if and only if there is an automorphism $\varphi \in \mathrm{Aut}(K[[x, y]])$ and a unit $u \in K[[x, y]]^*$ such that $f = u \cdot \varphi(g)$, and we denote this by $f \sim_c g$. The idea here is that φ and u still induce an isomorphism of the zero fibers of f and g.*

Definition 2.2. *We say that f is* right k-determined *if f is right equivalent to every $g \in K[[x, y]]$ whose k-jet coincides with that of f, where the k-jet of f is*

$$\mathrm{jet}_k(f) = f \mod \mathfrak{m}^{k+1} \in K[[x, y]]/\mathfrak{m}^{k+1}$$

the residue class of f modulo the $(k+1)$-st power of the maximal ideal. I.e., f is right k-determined if it is right equivalent to every g which coincides with f up to order k.

Similarly, we call f contact k-determined *if f is contact equivalent to every g whose k-jet coincides with that of f.*

In both situations, we say that f is finitely determined *if it is k-determined for some positive integer k, and we call the least such k the* determinacy *of f.*

Computing the determinacy of a dense polynomial can be expensive in characteristic 0. We want to use computations over positive chatacteistic to bound the determinacy. For this, we use known relations between the determinacy and the Milnor and Tjurina numbers.

Lemma 2.3 (page 64 in [2]). *For arbitrary fields of characteristic zero the right determinacy is at most $\mu(f)+1$ and the contact determinacy is at most $\tau(f) + 1$. For arbitrary characteristic, $2\mu(f)$ respectively $2\tau(f)$ are bounds for the degree of right respectively contact determinacy.*

Now we use the following result to bound the Milnor and Tjurina numbers in characteristic 0 by computations in characteristic p.

Lemma 2.4 (Corollary 3.1 in [4])**.** *Let $F \in \mathbb{Z}[x]$, $p \in \mathbb{Z}$ a prime number, and denote by F_p the image of F in $\mathbb{F}_p[[x]]$ and by F_0 the image of F in $\mathbb{Q}[[x]]$.*

If $\mu(F_p)$ is finite, then $\mu(F_0) \leq \mu(F_p)$. In particular, if $\mu(F_p)$ is finite for some p, then $\mu(F_0)$ is finite. The same holds for the Tjurina number.

Example 2.5. For $f = y^3 - x^2 \in \mathbb{C}[x, y]$, $\mu(f) = 2$ For $p = 2$ and $p = 3$, $\mu(F_p) = \infty$ and $\mu(F_5) = 2$.

This allows for the following strategy to compute the integral basis of a plane curve:

(1) Choose any small prime number, for example $p = 2$. Compute the Milnor number of f in $\mathbb{Z}_p[[x, y]]$. If the Milnor number is finite, then $\mu(f) + 1$ is a bound for the determinacy. If it is infinite, repeat with another prime p.
(2) If we truncate the polynomial f by standard degree at the determinacy, the characteristic exponents of f will not change, because they are invariant under right equivalence (see for example [3, Section 5.2].
(3) The integrality exponent of f can be computed from the characteristic exponents by a simple formula, so we can compute very fast the integrality exponent.
(4) We will derive bounds for truncating the original polynomial f in x-degree and y-degree based on the x-integrality exponent and the y-integrality exponent, such that the resulting polynomial has exactly the same integral basis generators.
(5) We truncate the polynomial using those bounds and compute the integral basis for the truncated polynomial.

Remark: the coefficients of the Puiseux expansions are not in general preserver by right equivalence. In our case we are truncating polynomials, which is a very special case of right equivalence. We have not been able to prove that truncating at the determinacy the singular part of the Puiseux expansions is not changed, but in all examples we tried, this is verified.

3. Known results about precision of factorization and computation of Puiseux expansions

We recall first two results that will be useful for our method.

The first result is about factorization over $K[[X]][Y]$.

Proposition 3.1. *(See* [5, Proposition 23]*.) Let $F \in K[[X]][Y]$, separable of degree d. Suppose given a modular factorisation*

$$F \equiv F_0 F_1 \ldots F_s \mod X^{N_0},\ N_0 > 2\kappa \tag{1}$$

where $F_0 \in K[[X]][Y]$ is a unit and for all $i \geq 1$ either F_i or its reciprocal polynomial $\tilde{F}_i$ is monic, and

$$\kappa = \kappa(F_0, ..., F_s) := \max_{I,J} \kappa(F_I, F_J),$$

the maximum of the lifting orders being taken over all disjoint subsets $I, J \subset 0, \ldots, s$, with $F_I = \prod_{i \in I} F_i$. Then there exists uniquely determined analytic factors $F_0^*, F_1^*, \ldots, F_k^*$ such that $F = F_0^* F_1^* \cdots F_k^*$, where

$$F_i^* \equiv F_i \mod X^{N_0 - \kappa}.$$

Moreover, starting from (1), we can compute the F_i^* up to any precision $N \geq N_0 - \kappa$ in $\mathcal{O}(dN)$ operations over K.

The authors provide the bound $\kappa \leq \delta/2$, where $\delta = \upsilon_X(\mathrm{Res}_Y(F, F_Y)) = 2\mathrm{IntExp}_X$, so $\kappa \leq \mathrm{IntExp}_X$.

The second result provides bound for the precision required for computing Puiseux expansions. We use this algorithm taking $\mathbb{K}_P = \mathbb{Q}$.

We denote by $(R_i)_{1 \leq i \leq \rho}$ the rational Puiseux expansions of H. To any R_i, we associate the following notations:

- e_i is the ramification index (equal to the largest denominator or the number of expansions in the branch),
- f_i is the residual degree,
- the *regularity index* r_i of a Puiseux series R_i of F with ramification index e_i is the least integer $N \geq \min(0, e\upsilon_X(R_i))$ such that, if

$$\lceil R_i \rceil^{N/e_i} = \lceil R' \rceil^{N/e_i}$$

  for some Puiseux series R' of F, then $R_i = R'$. We call $\lceil R_i \rceil^{r_i/e_i}$ the *singular part* of R_i in F.
- we define $v_i \in \mathbb{Q}$ as $\upsilon_X(F_Y(S))$, for any Puiseux series S associated to R_i.

  If H has only only one branch, then this number is the integrality exponent. Proof: if $F_i = (Y - \gamma_1) \ldots (Y - \gamma_r)$ then $(F_i)_Y(\gamma_1) = (1)(\gamma_1 - \gamma_2) \cdots (\gamma_1 - \gamma_{e_i}) + 0 + 0 + \cdots + 0$. The order of the product is equal to IntExp_X (it is the same formula).

  If H has many branches, then v_i can be bigger than the integrality exponent.

Algorithm 1 Half-RNP(H, P, n, π)

1: **Input:** $P \in \mathbb{K}[Z]$ irreducible; $H \in \mathbb{K}_P[X, Y]$ separable and monic in Y, with $d := \deg_Y(H) > 0$; $n \in \mathbb{N}$ (truncation order); π the current truncated parametrisation (initial call: $P = Z$, $\pi = (X, Y)$)

2: **Output:** All RPEs R_i of H such that $n - v_i \geq r_i$, with precision $(n - v_i)/e_i \geq r_i/e_i$

There seems to be an error in the bounds here, which don't agree with the following theorem in that same paper:

Lemma 3.2. ([5, Lemma 3.19]) *Let $n_0 \in \mathbb{N}$. To compute the RPE R_i with certified precision $n_0 > \frac{r_i}{e_i}$, it is necessary and sufficient to run* Half-RNP *with truncation bound $n = n_0 + v_i$. In particular, to ensure the computation of the singular part of R_i, it is necessary and sufficient to use a truncation bound $n > N_i$.*

We use the bounds in the Lemma, which have been confirmed by the authors and by computational experiments.

To give general bounds for the precision n required we define a new number:

$$\alpha = \sum r_i,$$

the sum taken over all Puiseux branches. This number is bigger than the integrality exponent but not much bigger. For example, if we have only one branch with seven Puiseux expansions and one characteristic exponent $1/7$, then $\alpha = 1$ and the integrality exponent is $6/7$.

We have:

- $r_i \leq \alpha$ (clear by definition)
- $v_i \leq \mathrm{IntExp}_X \leq \alpha$ when there is only one branch, but not in general.

Based on the last observation, we define $\tilde{v}_i = v_X((F_i)_Y(S))$, for any Puiseux series S associated to R_i. This number will be used when we apply Algorithm 1 to only one branch of F.

4. Local contribution to the integral basis at the origin

In this section we focus on the computation of the local contribution at the origin (that is, we want to keep the unit too).

The steps are now as follows:

(1) Proposition 3.1 says that if we truncate F at precision n in X and we factorize the truncated polynomial $\bar{F}$ in $K[[X]][Y]$, we will get the factors of F with precision $n - \kappa$.
(2) We are interested in computing the factors with precision at least equal to the integrality exponent in X of F, we call this number IntExp_X. That is, we need $n - \kappa \geq \mathrm{IntExp}_X$.
(3) In that paper the authors provide the bound $\kappa \leq \delta/2$, where $\delta = v_X\left(\mathrm{Res}_Y(F, F_Y)\right) = 2\mathrm{IntExp}_X$, so $\kappa \leq \mathrm{IntExp}_X$ and it is enough for us to consider $n \geq 2\mathrm{IntExp}_X$.
(4) For the computation of the integral basis we also need the singular part of the Puiseux expansions. If we want to compute the Puiseux expansions from the factors $F_1, \ldots, F_s$, we would need to compute the factors with precision higher than the integrality exponent.

    However if we compute the Puiseux expansions from $\bar{F}$, the same proposition 3.1 says that we will obtain the Puiseux expansions with precision at least IntExp_X, and hence the singular part will be correctly computed.
(5) I am not sure about this last step. In any case, we can compute the singular part of the Puiseux expansions by truncating at $3 \times \mathrm{IntExp}_X$. The argument is as follows:
    (a) We apply Lemma 3.2 to one factor F_i.
    (b) In this case $v_i = v_X((F_i)_Y)$ and this is smaller than IntExp_X (the integrality exponent of one branch is smaller than that of the full polynomial).
    (c) By [5, Lemma 3.21], $r_i/e_i \leq v_i$.

(d) By Lemma 3.2, we need $n \geq v_i + r_i/e_i$, and $v_i + r_i/e_i \leq 2\mathrm{IntExp}_X$. That is, we need the factor F_i developed to degree $2\mathrm{IntExp}_X$ and by Proposition 3.1 we can get this precision by computing a factorization of F modulo $3\mathrm{IntExp}_X$.

(6) Putting everything together, we can truncate the powers of X of F at order $3\mathrm{IntExp}_X$ and we will be able to compute the correct factors up to the integrality exponent and the singular part of the Puiseux expansions and hence we will compute the correct integral basis.

We obtain the following result.

Proposition 4.1. *Let $F \in K[X,Y]$ be monic in Y, let κ be the maximum of the lifting orders as defined in Proposition 3.1. Let $N_0 \in \mathbb{N}$, $N_0 > 3\mathrm{IntExp}_X$, where IntExp_X is the integrality exponent of F in X (the minimum exponent of a power of X in the conductor of F). Let $\overline{F} \in K[X,Y]$ be such that $F \equiv \overline{F} \mod X^{N_0}$. Then the generators of the local contribution at the origin to the integral basis of $\overline{F}$ are also generators of the local contribution at the origin to the integral basis of F. That is, for computing the local contribution at the origin to the integral basis of F we can truncate the powers of X of F above degree $3\mathrm{IntExp}_X$.*

Proof. Let $F = F_0^* F_1^* \dots F_k^*$ be the analytic factorization of F into branches. To compute the local contribution at the original for the integral basis of F, we need to compute these factors developed up to the integrality exponent, and the singular part of the Puiseux expansions of F at the origin.

Let κ be the maximum of the lifting orders as defined in Proposition 3.1. We know that $\kappa \leq \delta/2$, where $\delta = v_X(\mathrm{Res}_Y(F, F_Y)) = 2\mathrm{IntExp}_X$, so $\kappa \leq \mathrm{IntExp}_X$.

Since $N_0 > 3\mathrm{IntExp}_X$, in particular $N_0 > 2\kappa$ and $N_0 - \kappa > 2\mathrm{IntExp}_X$ and hence by Proposition 3.1, if we compute a factorization

$$F \equiv F_0 F_1 \dots F_s \mod X^{N_0},$$

then $F_i^* \equiv F_i \mod X^{2\mathrm{IntExp}_X}$.

Finally, we follow Proposition 3.2 to compute the singular part of the Puiseux expansions of F_i, $1 \leq i \leq k$, we need the factor F_i computed with precision at least $\frac{r_i}{e_i} + v_i$. We have shown that $\frac{r_i}{e_i} + v_i \leq 2\mathrm{IntExp}_X$, so the result follows. □

5. Local integral basis at the origin

When we are interested in the local integral basis at the origin we can also truncate powers of Y.

One possible approach is the following:

(1) Applying again [5, Proposition 23] looking at the polynomial in $K[[Y]][X]$ with high enough precision, we obtain that the factors at the origin are uniquely determined and they are the same factors as the factors obtained when we develop the factorization in terms of X (but with a different parametrization), so we are able to recover the correct information.

(2) The factor g_0 however corresponds to the branches of f at $Y = 0$ and the factor f_0 corresponds to the branches of f at $X = 0$, so

we cannot recover the Puiseux expansions of f_0 from the Puiseux expansions of g_0.

(3) Hence when we apply a truncation in Y we can expect to compute correctly the local integral basis but not the local contribution.

We now formalize these ideas and calculate the required precision. For this, we need to be able to go from Puiseux expansions in X to Puiseux expansions in Y. This can be done using inversion formulas, see for example [1].

(1) We start with a polynomial $F \in K[X,Y]$ with analytic factorization in $K[[X]][Y]$:
$$F = F_0^* F_1^* \dots F_s^*,$$
where $F_k^* \in K[[X]][Y]$ and F_0^* is the unit corresponding to the branches outside the origin). Let d_k be the degree in Y of F_k^* and let e_k be the ramification index of F_k^* (the largest denominator of the Puiseux expansions).

(2) If we have a factorization
$$F \equiv F_0 F_1 \dots F_s \mod X^{N_0}, \tag{2}$$
then by Proposition 3.1, $F_k \equiv F_k^* \mod X^{N_0-\kappa}$.

(3) Now we truncate in Y. Let $\overline{F}^Y$ be the truncation of F modulo Y^{M_0}, for some value M_0 to determine. We can factorize this polynomial in $K[[X]][Y]$ as
$$\overline{F}^Y \equiv H_0 H_1 \dots H_s \mod X^{N_0},$$
and we want to know if $H_k \equiv F_k^* \mod X^{N_0-\kappa}$ for $k > 0$ (we cannot expect $H_0 \equiv F_0^* \mod X^{N_0-\kappa}$).

(4) We can also factorize $\overline{F}^Y$ in $K[[Y]][X]$:
$$\overline{F}^Y \equiv G_0 G_1 \dots G_s \mod Y^{M_0}.$$

(5) Applying Proposition 3.1 to F in $K[[Y]][X]$ we obtain that
$$G_k \equiv G_k^* \mod Y^{M_0-\kappa_Y}$$
where G_k^* are the analytic factors of F in $K[[Y]][X]$.

By the same argument G_k agrees with the analytic factors of $\overline{F}^Y$ mod $Y^{M_0-\kappa_Y}$. Hence: the analytic factors in $K[[Y]][X]$ of F and $\overline{F}^Y$ are the same modulo $Y^{M_0-\kappa_Y}$.

The question is what precision do we get when we go from these factors to factors in $K[[X]][Y]$. We need to compute the factors correctly up to degree IntExp_X in X, and we study now up to which degree we can truncate in Y.

(1) Let $\overline{F}^Y$ be the truncation of F at some degree N_0 in Y. We want to know a good K.

(2) Assume $K \geq 3\mathrm{IntExp}_Y$.

(3) We can factorize $\overline{F}^Y$ in $K[[Y]][X]$:
$$\overline{F}^Y \equiv G_0 G_1 \dots G_s \mod Y^{N_0}.$$

(4) By Proposition 3.1, since $K \geq 2\mathrm{IntExp}_Y$, the factors are correct up to degree $n = N_0 - \mathrm{IntExp}_Y$. That is,
$$G_k \equiv G_k^* \mod Y^{N_0 - \mathrm{IntExp}_Y}$$
where G_k^*, $0 \leq k \leq s$, are the analytic factors of F in $K[[Y]][X]$.

(5) Consider the k-th branch. Let ϵ_k be the ramification index of G_k^* (the largest denominator in the expansions), $\tilde{v}_k = v_Y((G_k^*(\eta(Y^{1/\epsilon_k})))$ and r_k the regularity index (so that the singular part goes up to degree r_k/ϵ_k). Hence
$$G_k^* = \prod_{i=1}^{\epsilon_k}(X - \eta(\xi^i Y^{1/\epsilon_k})).$$
We have
$$n - \tilde{v}_k \geq n - \mathrm{IntExp}_Y = N_0 - 2\mathrm{IntExp}_Y \geq \mathrm{IntExp}_Y \geq \tilde{v}_k \geq \frac{r_k}{\epsilon_k}.$$
Then by Lemma 3.2, since the polynomial G_k^* is computed with precision bound $n = N_0 - \mathrm{IntExp}_Y$, we obtain the Puiseux expansions with precision $n_0 = n - \tilde{v}_k \geq \mathrm{IntExp}_Y$. Let $D = n - \tilde{v}_k$.

(6) The polynomial $\eta(U)$ corresponding to the k-th branch is correct up to degree $D\epsilon_k$: the Puiseux expansion $\eta(Y^{1/\epsilon_k})$ is correct to degree D and we take $U = Y^{1/\epsilon_k}$.

(7) Applying the inverse formula to the Puiseux expansions we get a polynomial $\gamma(T)$ which is correct up to the same degree $D\epsilon_k$ in T. Now we replace $T = X^{1/e_k}$ to get the Puiseux expansions in X, $\gamma(X^{1/e_k})$. We get correct Puiseux expansions up to degree $D\epsilon_k/e_k$.

(8) As a conclusion, if we have a factorization
$$F \equiv F_0 F_1 \ldots F_s \mod X^{N_0}, \tag{3}$$
then by Proposition 3.1, $F_k \equiv F_k^* \mod X^{N_0 - \kappa}$, where $F_k^* \in K[[X]][Y]$ are the analytic factors at the origin and F_0^* is the unit corresponding to the branches outside the origin.

(9) We need this number $D\epsilon_k/e_k$ to be bigger than IntExp_X. So we get $D \geq \frac{e_k}{\epsilon_k}\mathrm{IntExp}_X$ and then
$$N_0 - \mathrm{IntExp}_Y - \tilde{v}_k \geq \frac{e_k}{\epsilon_k}\mathrm{IntExp}_Y.$$
So we get
$$N_0 \geq \frac{e_k}{\epsilon_k}\mathrm{IntExp}_X + \mathrm{IntExp}_Y + \tilde{v}_k.$$

(10) Consider $\frac{e}{\epsilon}$ the largest $\frac{e_k}{\epsilon_k}$ over all branches. So we need
$$N_0 > \frac{e}{\epsilon}\mathrm{IntExp}_X + 2\mathrm{IntExp}_Y,$$
which is the bound we were looking for.

We obtain the following result

Proposition 5.1. *Let $F \in K[X, Y]$ be monic in Y. Assume $F = F_0^* F_1^* \ldots F_s^*$ is the analytic factorization of F, where $F_k^* \in K[[X]][Y]$ and F_0^* is the unit corresponding to the branches outside the origin. Let d_k be the degree in Y of F_k^* and let e_k be the ramification index of F_k^* (the largest denominator of* *the Puiseux expansions). Let $F = G_0^* G_1^* \dots G_s^*$ be the analytic factorization of F in $K[[Y]][X]$ and let ϵ_k be the ramification index of G_k^*. Let $M_0 \in \mathbb{N}$, $M_0 > \frac{e_k}{\epsilon_k}\mathrm{IntExp}_X + 2\mathrm{IntExp}_Y$ for all $1 \leq k \leq s$ and $M_0 > 3\mathrm{IntExp}_Y$. Let $\overline{F} \in K[X,Y]$ be such that $F \equiv \overline{F} \mod Y^{M_0}$. Then the generators of the local integral basis of $\overline{F}$ at the origin are also generators of the local integral basis of F at the origin. That is, for computing the local the integral basis of F at the origin we can truncate the powers of X of F at degree M_0.*

Proof. Let $F = G_0 G_1 \dots G_s$ be a factorization of F modulo Y^{M_0}. By Proposition 3.1, we know that $G_i \equiv G_i^\star \mod Y^{M_0 - \mathrm{IntExp}_Y}$. Following 3.2 (as in the proof of Proposition 4.1), if we compute the Puiseux expansions of G_i, they will agree with the Puiseux expansions of G_i^* up to degree at least $\frac{e}{\epsilon}\mathrm{IntExp}_X$, where $\frac{e}{\epsilon}$ is the largest possible $\frac{e_k}{\epsilon_k}$, $1 \leq k \leq s$.

If we apply the inversion formula to these expansions we will get the Puiseux expansions of F_k in terms of X correct up to degree $\frac{\epsilon_k}{e_k}\frac{e}{\epsilon}\mathrm{IntExp}_X \geq \mathrm{IntExp}_X$. Hence we can also compute the correct factors $F_0^*, F_1^*, \dots, F_s^*$ up to degree IntExp_X, as wanted. □

6. Timings

We get very good timings.

References

[1] Evelia Barroso, Pedro Pérez, and Patrick Popescu-Pampu. Variations on inversion theorems for newton-puiseux series. *Mathematische Annalen*, 368, 08 2017.

[2] Yousra Boubakri, Gert-Martin Greuel, and Thomas Markwig. Invariants of hypersurface singularities in positive characteristic. *Revista Matemática Complutense*, 25(1):61–85, 2012.

[3] Theo De Jong and Gerhard Pfister. *Local analytic geometry: Basic theory and applications*. Springer Science & Business Media, 2013.

[4] Gert-Martin Greuel and Gerhard Pfister. Semicontinuity of singularity invariants in families of formal power series. In Javier Fernández de Bobadilla, Tamás László, and András Stipsicz, editors, *Singularities and Their Interaction with Geometry and Low Dimensional Topology*, pages 207–245, Cham, 2021. Springer International Publishing.

[5] Adrien Poteaux and Martin Weimann. Computing puiseux series: a fast divide and conquer algorithm. *Annales Henri Lebesgue*, 4:1061–1102, 2021.